

When and What to Speak: An Autonomous Embodied Chat Agent in Symmetrical Reality

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ABSTRACT

Most LLM dialog agents respond only when prompted. We study an autonomous inserter for multi-party talk that decides whether and when to join. Insertion is modeled along three dimensions: speaker persona, scene context, and topic. Using a Qwen-7B backbone, we train persona-, scene-, and topic-aware regression heads to predict a continuous willingness score, plus a classifier over three intervention styles. Deployed in a Unity multi-party simulation, the agent monitors dialogue and inserts itself accordingly. Three-stage evaluations on transcripts, dyadic chats, and group discussions compare model and human insertion judgments. This provides an measure of how human-like its timing can be.

Index Terms: Intelligent agents, Natural language processing, Machine learning models

1 INTRODUCTION

In current human–computer dialog systems, most language model based agents still behave as passive responders. They start to generate output only when the user explicitly speaks, issues a command, or triggers a wake word, and otherwise remain silent. This interaction pattern still follows a command and response paradigm, where the agent executes clearly specified requests instead of making proactive decisions grounded in its own understanding of the situation.

In this work we focus on how to let a virtual agent display genuine autonomy in multi party conversation, especially in terms of insertion timing and willingness to participate. Rather than designing the agent as a one to one chat partner, we place it in an ongoing group conversation as a latent participant that is always present. The agent continuously observes information about speaker persona, surrounding scene, and conversational topic, and uses these signals to estimate whether the group is open to help from an AI, whether the current turn leaves a natural opening, and whether its contribution would meaningfully support or advance the discussion. Only when these conditions are satisfied does the agent decide to speak. By explicitly treating whether to speak and when to speak as decision variables, the system allocates the agent a concrete space of choice, and its autonomy becomes observable through the timing and manner of its interventions.

2 METHOD

Module 1: Persona-Aware Insertion Willingness

In Module 1, we train the agent to estimate how interruptible a speaker is given their background and current utterance. We first collect and clean publicly available data with occupational and personality tags, and construct a corpus of about **300K** annotated persona–utterance instances. For each persona, we author multiple plausible utterances and assign a continuous willingness score in

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Figure 1: Our system implements an autonomous conversational agent for multi-party dialogue that continuously senses speaker persona, scene context, and topic, and then decides whether to speak, when to speak, and what kind of response to give.

[0.1, 0.9], where 0.1 denotes a strong opportunity for interruption and 0.9 indicates that interruption is not appropriate. To increase coverage, we additionally use a large language model to generate diverse utterances for the same persona under different target willingness levels.

Based on this corpus, we fine-tune a Qwen-7B model with a regression head that takes the persona description and utterance as input and outputs the predicted willingness. On our annotated dataset, the mean absolute error (MAE) drops from about 0.23 for the base model to around 0.02 after fine-tuning, while an off-the-shelf *ChatGPT-4o mini* model yields an MAE of roughly 0.30 on the same data. This indicates that the fine-tuned regressor can accurately predict insertion willingness and provide a reliable signal for downstream decision modules that control whether and when the agent should step into the conversation.

Module 2: Scene-Aware Insertion Willingness

While Module 1 focuses on who the speaker is, Module 2 models how the surrounding scene shapes the agent’s willingness to join an ongoing conversation. We construct a structured scene descriptor with six dimensions: (1) *relationship*, describing how the participants are related (for example, family, close friends, classmates, colleagues, or strangers); (2) *preference*, indicating whether the group is receptive to AI assistance or prefers to keep the conversation private; (3) *sensitivity*, reflecting how sensitive the topic or participants are to AI involvement; (4) *group size*, specifying how many people are present; (5) *formality*, representing how formal



Figure 2: Across everyday group settings, the autonomous agent reads the scene, topic, and personalities to act less like a passive tool and more like a considerate group member.

the setting is; and (6) *location type* and *time*, describing where and when the interaction takes place.

Each dimension is discretized into categories, and we assign a weight to each category to reflect how strongly it should encourage or discourage agent intervention. By combining these dimension values and applying the weighting scheme, we generate natural language scene descriptions together with a continuous insertion willingness label in $[0.1, 0.9]$, where lower values indicate that the scene invites agent participation and higher values indicate that interruption is less appropriate. This process yields about **300K** annotated scene instances.

Building on the same Qwen-7B backbone, we attach a scene regression head and fine-tune it to map the scene description to the corresponding willingness score. On our annotated dataset, the mean absolute error (MAE) of the base model is about 0.26, whereas the fine-tuned scene head reduces the MAE to approximately 0.0065. For comparison, an off-the-shelf *ChatGPT-4o mini* model achieves an MAE of 0.1851 under the same protocol. These results suggest that the fine-tuned scene regressor can accurately interpret contextual cues and provide a reliable prior about how willing the agent should be to speak in a given situation.

Module 3: Topic-Aware Content and Strategy Modeling

Module 3 focuses on what is being talked about. Here we train the agent to infer its willingness to join a conversation directly from the dialogue content and its underlying topic. We retrieve and cluster large-scale conversational texts from the web that cover everyday themes such as work, study, family, entertainment, and health. For each topic cluster, we construct multi-turn dialogues and annotate (1) which utterances are suitable insertion points, and (2) a continuous willingness score for the agent after each user utterance. Given both the topic description and the utterance content, the agent can estimate how strongly it should consider entering the dialogue at that point.

This topic-centered pipeline yields about **450K** annotated dialogue instances, each paired with a topic label, local dialogue context, and an insertion willingness score. We attach a topic regression head on top of the Qwen-7B backbone and fine-tune it to map the combination of topic description and current utterance to the corresponding willingness value. On our annotated dataset, the MAE decreases from about 0.21 for the base model to approximately 0.08 after fine-tuning, indicating that the model can reliably use content and topic cues to judge when it is appropriate to speak.

Beyond continuous willingness estimation, we also model *how* the agent should respond. We categorize the agent’s intended be-

havior at intervention points into three high-level strategy types: *risk_alert*, *rational_analysis*, and *comforting*. When the user expresses signals such as anxiety, confusion, or emotional distress, the agent is trained to favor the corresponding strategy. We implement a discrete classification head for these three strategies and fine-tune it on the annotated interventions, improving strategy prediction accuracy on the test set from 74% to 94%. This enables the agent not only to decide whether to

3 EXPERIMENT DESIGN

We design a three stage experiment to evaluate when and how the agent inserts itself into human conversations. In Stage 1, we use complete multi turn dialogue transcripts with human annotated insertion points. We replay each transcript turn by turn to the agent, log its willingness score over time, and mark the turns where it chooses to insert. We then compare the agent’s insertion turns with the human annotations to quantify alignment and summarize typical matches and mismatches. In Stage 2, two participants conduct a live conversation while the agent listens in real time and decides its own moments and strategies to speak. After the session, participants review each insertion event with its local context and rate whether the timing and intervention are appropriate. We aggregate these event level ratings and analyze how they relate to the agent’s insertion decisions. In Stage 3, the agent joins live multi party conversations using the same autonomous listening and insertion procedure. After the discussion, all participants give overall ratings of how well and how reasonably the agent inserted itself into the group dialogue, and provide brief feedback on whether the insertions supported or disrupted the group flow.

4 CONCLUSION AND FUTURE WORK

In this work we operationalize autonomous insertion in multi party conversation by factoring the decision of whether and when to speak into three components: who is talking, in what kind of scene, and what is being discussed. We train persona aware, scene aware, and topic aware regression heads on top of a Qwen 7B backbone, together with a discrete strategy classifier, to estimate both the agent’s willingness to intervene and the style in which it should do so, and deploy them in a Unity based multi party simulation where the agent monitors ongoing dialogue, integrates these signals, and decides when and how to join as an additional participant, with experiments spanning static transcripts, live dyadic conversation, and real time group discussion.

Our current estimators rely on text descriptions of persona, scene, and topic and are trained mostly offline, so extending them to multimodal cues, unifying the three modules into a policy refined through online interaction or reinforcement learning, and supporting richer user adaptive intervention styles while explicitly addressing consent, privacy, and transparency when an AI agent joins group discussions could better align insertion behavior with user preferences.

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